CRF, SVM and Bi-LSTM CRF Trained on Twitter Only (WNUT17): What Raised the F1 and What Did Not

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Abstract

WNUT17 tests whether named entity recognition (NER) systems can find entities they have never seen. We train a CRF, a class-weighted SVM and a Bi-LSTM CRF on the **Twitter training file only** (no YouTube or Reddit data in training), with dev used only to choose settings and stop training, seed 42 for every run, and one final scoring on the full gold test file (1,287 sequences, 1,079 entities). The baseline Bi-LSTM CRF (frozen GloVe Twitter vectors, character CNN, capitalization embedding) scores **40.1** entity F1. We tried to raise it in five ways. Word dropout, fine-tuned embeddings, a larger hidden layer and POS-tag input each looked better on dev (+0.4 to +1.9 F1) but none raised the test F1 (-2.2 to -0.3), and a precision-recall shift at decoding tuned on dev lowered it; the variant chosen by dev scored 37.9 on test. An ensemble of the six seed-42 models, defined in advance, was the only change that rose on both dev and test: **41.5** (+1.4; paired bootstrap interval -0.2 to 2.9), close to the 41.6 obtained earlier when YouTube data were added to training. The SVM reaches 29.8 and the CRF 16.8. Recall on unseen entities is only 11.2 to 30.6%, and corporation, product, creative-work and group reach at most 26.1 F1 for every model.

Keywords: named entity recognition, WNUT17, emerging entities, CRF, SVM, Bi-LSTM CRF, GloVe, word dropout, noisy text

1. Introduction

Named entity recognition (NER) finds spans of text that name people, places, organizations and other things, and labels their type. Systems trained on news do well because entities repeat between training and test. Social media breaks this: text is short, spelling and capitalization are unreliable, and new names appear constantly.

WNUT17 (Derczynski et al., 2017) was built to test exactly this. Entities in dev and test were chosen so that they do not appear in the training tweets, and the best of seven shared-task systems reached only 41.86 entity F1. The assignment asks for a CRF, an SVM and a Bi-LSTM CRF to be implemented and compared. Because no Indonesian dataset was assigned to our group, all work uses the English WNUT17 data.

This report differs from our earlier run in two ways. First, training uses **Twitter data only**: the YouTube dev file is no longer added to the training data, so the models must generalize from tweets to YouTube, Reddit and StackExchange text. Second, only seed 42 is used, and we try to improve the best model. Because the earlier run trained on train plus dev, its 41.6 F1 is not the right starting point: the comparable baseline is the Twitter-only Bi-LSTM CRF at 40.1, a gap of 1.5 points to close.

Contributions. (1) One evaluation protocol for three models, with full-test-set scoring and strict entity F1. (2) A step-by-step attempt to raise the Bi-LSTM CRF F1, with every choice made on dev, including the changes that did not carry over to test. (3) Bootstrap intervals, ablations, an error analysis with a confusion matrix per model, and examples. (4) A complete list of deviations from the Stage 1 plan, including preprocessing bugs found and fixed.

2. Related Work

NER on social media. Ritter et al. (2011) showed that tools trained on news lose much accuracy on tweets and built a tweet-specific tagger. The WNUT17 shared task (Derczynski et al., 2017) focused on *novel and emerging* entities: the organizers removed from dev and test every text containing an entity seen in training. Seven teams took part, from feature-based CRFs to neural models, and the best, UH-RiTUAL (Aguilar et al., 2017), used a multi-task neural network.

Feature-based models. Linear-chain CRFs (Lafferty et al., 2001) became the standard for NER with hand-built features and lexicons (McCallum and Li, 2003). Support vector machines were applied to chunking and NER by classifying each token independently over a window of features (Kudo and Matsumoto, 2001; Isozaki and Kazawa, 2002).

Neural sequence labeling. Bi-LSTM encoders with a CRF output layer (Huang et al., 2015) removed most feature engineering; character-level CNNs or LSTMs (Chiu and Nichols, 2016; Lample et al., 2016; Ma and Hovy, 2016) help with spelling variants and unseen words. Pretrained vectors such as GloVe (Pennington et al., 2014), trained on tweets

in the Twitter release, give the model knowledge of words absent from the labeled data. Dropout (Srivastava et al., 2014) is the standard regularizer, and replacing rare training words by an unknown token (Lample et al., 2016) teaches a model to cope with words it has not seen; our word dropout follows the same idea. Contextual encoders such as BERT (Devlin et al., 2019), BERTweet (Nguyen et al., 2020) and contextual string embeddings (Akbik et al., 2018) go further but are outside this assignment, which fixes the three model families.

3. Methodology

3.1 Data and protocol (Twitter-only training)

We use the three official files of github.com/leondz/emerging_entities_17 without re-splitting: **wnut17train.conll** (Twitter 2010), **emerging.dev.conll** (YouTube comments) and **emerging.test.annotated** (Reddit, StackExchange, Twitter 2017). The test set is the **entire** gold file; the raw file *emerging.test.conll* is not used. Labels are IOB2 with six types (13 labels).

Table 1. Dataset statistics recomputed from the released files (all Stage 1 counts reproduce).

	Train (Twitter)	Dev (YouTube)	Test (Reddit, SE, Twitter 2017)
Role in this report	the only training data	choose settings, stop training	final scoring, once
Sequences	3,394	1,009	1,287
Tokens	62,730	15,733	23,394
Entities	1,975	836	1,079
Tokens labeled O	95.0%	92.1%	92.6%
Entities unseen in train (mention level)	-	96.3%	97.8%
Entity tokens out of vocabulary (vs train)	-	56.0%	58.3%

Protocol. (1) Train every model on the train file only; dev is used to select settings and to stop training, never to update weights. (2) Choose settings and improvements by dev entity F1 only. (3) Score the chosen models on the test file. Test scores of the intermediate improvement steps are printed for transparency but were never used to choose. (4) Seed 42 for every run. Metrics: strict entity precision, recall and F1 (seqeval, IOB2), the shared-task surface F1 and token accuracy. Uncertainty: 95% bootstrap intervals from 2,000 resamples of test sentences.

3.2 Preprocessing

- **Reading.** A blank line ends a document; in train, 2,394 tab-only lines end a sentence inside a tweet (3,394 sequences from 1,000 tweets). Tokens are used as released, so labels stay aligned; the BIO check finds 0 violations.
- **Normalized copy of each token** (for lookup and lexical features; the original token is kept for shape features): HTML unescape, NFKC, URL (including bare domains) to <URL>, @user to <USER>, hashtag body without #, every digit to 0, repeated characters capped at three (digit runs are not capped), lowercase. No stemming, lemmatization or stop-word removal.
- **GloVe Twitter 200d** lookup with ordered keys (lowercase token, normalized form, special keys, hashtag body). Coverage: 93.4% of train tokens, 96.7% of test tokens; for entity tokens 94.2% in train but only 84.9% in test. Tokens without a vector get zeros.
- **CRF and SVM features** (window +-2): lexical, context words and bigrams, prefixes and suffixes, shape, social-media flags, NLTK POS tags, K-Means cluster IDs (k = 500) of GloVe vectors; the SVM also gets the token's GloVe vector (x 0.3) in a sparse matrix that drops features seen fewer than three times. Capitalization: shape and is_title features keep *Google* and *google* apart; everything lexical is lowercased.
- **Bi-LSTM CRF inputs:** frozen GloVe vector, a character-CNN vector (first 20 characters) and a 10-class capitalization embedding.

3.3 Models

CRF: linear-chain CRF (sklearn-crfsuite, L-BFGS, c1 = 0.3, c2 = 0.003, 300 iterations). **SVM**: one-vs-rest LinearSVC, C = 0.03, balanced class weights, no BIO repair. **Bi-LSTM CRF baseline**: one bidirectional LSTM layer (128 units per direction) over the 258-dimensional token representation, a linear layer and a CRF layer decoded with Viterbi; dropout 0.5, Adam 0.001, batch 32, early stopping on dev (patience 8, at most 35 epochs). It is written in JAX and optax because PyTorch could not be installed in the working environment; the CRF and Viterbi were checked against brute-force enumeration.

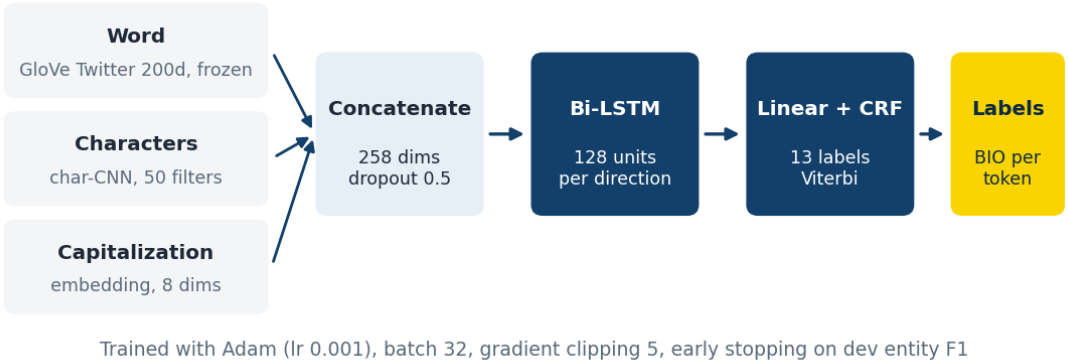


Figure 1. Bi-LSTM CRF baseline architecture.

3.4 Methods tried to raise the Bi-LSTM CRF F1

- **Word dropout (V1).** During training the GloVe vector of a random 10% of tokens is set to zero, exactly what happens to a word without a vector. In test, 15.1% of entity tokens have no vector, against 5.8% in train, so the baseline rarely learns to rely on characters and context when the word vector is missing.
- **Fine-tuned GloVe (V2).** The frozen vector table becomes trainable with an effective learning rate ten times smaller than the other weights, so the vectors can adapt to entity roles without drifting on 3,394 sentences; the vector of unknown words also becomes trainable.
- **Larger hidden layer (V3).** 256 units per direction give the encoder more capacity for context.
- **POS-tag input (V4).** The NLTK tag of each token enters as a 12-dimensional embedding. Proper-noun tags (NNP) flag names, and the Reddit part of the test set was itself selected with a proper-noun tagger.
- **Combination (V5).** Rule fixed in advance: keep every single change that gains at least 0.3 dev F1 and train them together.
- **Precision-recall shift at decoding.** A constant delta is subtracted from the score of the O tag before Viterbi decoding. It needs no retraining; delta is chosen on dev among 0, 0.5, ..., 3.0 (ties go to the smaller value).
- **Ensemble.** Defined before its scores were seen: average the emission scores and the CRF start, end and transition scores of the six seed-42 models (V0 to V5), then decode with Viterbi. No variant has to be chosen; the members differ in hidden size, embedding training, word dropout and POS input, not in the seed.

4. Experiments and Results

4.1 Search on dev (Twitter-only training)

Table 2. Search space and selected settings (CRF and SVM searches were run before the preprocessing fix, Appendix B).

Model	Searched	Selected	Dev F1
CRF	c1 in {0.1, 0.3, 1.0} x c2 in {0.003, 0.01, 0.03}, with and without cluster features (18 runs)	c1 = 0.3, c2 = 0.003, all features	24.5
SVM	C in {0.01, 0.03, 0.1, 0.3}; class weights; GloVe vector; BIO repair on or off	C = 0.03, balanced, GloVe, no repair	44.0
Bi-LSTM CRF	four single changes, their combination, an ensemble of the six models, and a decoding shift (Section 4.3)	baseline reported; dev choice among single models: V3; ensemble of V0 to V5	52.7 (ensemble)

4.2 Main results

Table 3. Test results in percent (models trained on the Twitter train file only, seed 42, scored once).

Model (train file only)	Precision	Recall	F1 entity	F1 surface	Token acc.	Earlier train+dev F1 (reference)
CRF	31.9	11.4	16.8	17.6	92.2	27.7
SVM	31.5	28.3	29.8	26.0	92.1	31.1
Bi-LSTM CRF (baseline)	57.2	30.9	40.1	39.0	94.2	42.5
Bi-LSTM CRF (ensemble)	**61.3**	**31.3**	**41.5**	**39.9**	**94.4**	**-**
Bi-LSTM CRF (V3 + shift, the single model chosen by dev)	45.8	32.2	37.9	36.6	93.8	-
Reference: predict O everywhere	-	-	-	-	92.6	-
Reference: best shared-task system	-	-	41.86	40.24	-	-

Bi-LSTM CRF baseline: 40.1; ensemble of six variants: **41.5** (+1.4); the single variant chosen by dev, with its dev-chosen shift: 37.9 (-2.2); SVM 29.8; CRF 16.8. The ensemble is 0.1 below the 41.6 of the earlier train+dev run (mean of three seeds; seed 42 alone gave 42.5) although it never sees YouTube data in training. Removing the dev data from training costs the CRF 10.9 F1 points and the SVM 1.3, and costs the baseline Bi-LSTM CRF 2.4 (seed 42 against seed 42). Token accuracy is close to the 92.6% of predicting O everywhere for the CRF and SVM, so entity F1 is the meaningful measure.

4.3 Trying to raise the Bi-LSTM CRF F1

Every change was trained on train only (seed 42) and stopped on dev. Table 5 lists dev F1 (used to choose) and test F1 (recorded, not used to choose). The dev gains did not carry over to test, and this is the main finding of this section.

Table 4. Everything tried on the Bi-LSTM CRF (seed 42, Twitter-only training). Changes are against V0.

Step	Change	Dev F1	Dev change	Test F1	Test change
V0	Baseline: frozen GloVe, char-CNN, capitalization, 128 units	51.5	-	40.1	-
V1	Word dropout 0.1 on the GloVe vector	52.3	+0.8	39.8	-0.3
V2	Fine-tuned GloVe (0.1 x learning rate)	52.0	+0.4	39.4	-0.7
V3	Hidden size 256	53.4	+1.9	37.9	-2.2
V4	POS-tag embedding (12 dims)	52.2	+0.7	39.6	-0.5
V5	All four changes together	52.7	+1.2	40.5	+0.4
E	**Ensemble of V0 to V5 (average scores, no shift)**	**52.7**	**+1.2**	**41.5**	**+1.4**
V0+	Baseline with shift delta 1.0 (dev choice)	54.1	+2.6	39.2	-0.9
V3+	V3 with shift delta 1.0 (single model chosen by dev)	54.3	+2.8	37.9	-2.2
E+	Ensemble with shift delta 1.5 (dev choice)	56.0	+4.4	40.4	+0.3

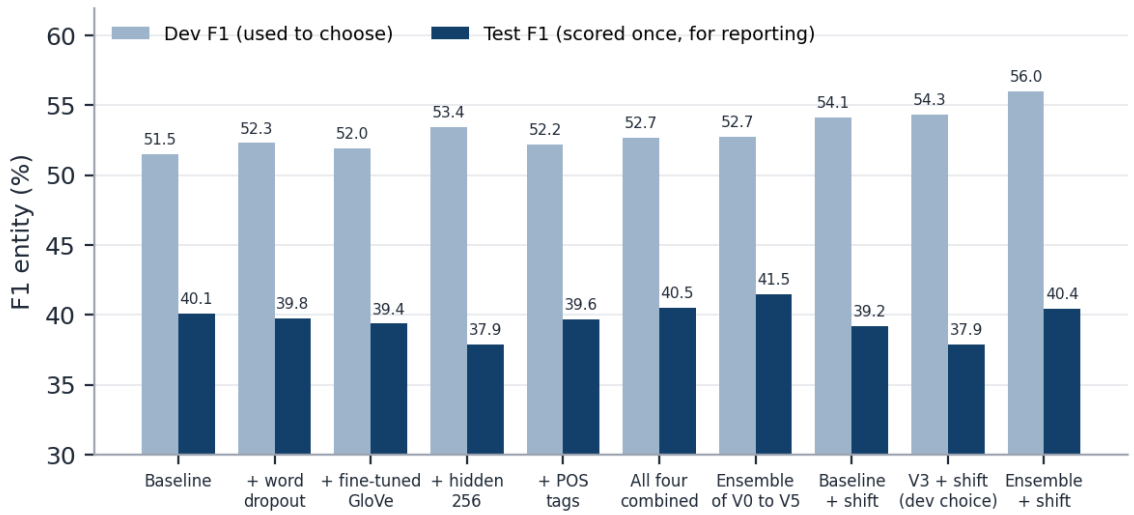


Figure 2. Dev F1 (light) and test F1 (dark) for every model tried. Only the dev bars were used to choose.

- **Word dropout** (V1). Aim: teach the model to use characters and context when the word vector is missing, which is common for test entities. Dev +0.8, test -0.3.
- **Fine-tuned GloVe** (V2). Aim: adapt the vectors to entity roles. Dev +0.4, test -0.7.
- **Hidden size 256** (V3). Aim: more capacity for context. It had the largest dev gain (+1.9) and the largest test loss (-2.2), so it is the clearest sign that dev choices did not transfer.
- **POS-tag input** (V4). Aim: an explicit proper-noun signal. Dev +0.7, test -0.5.
- **All four together** (V5). Dev +1.2, test +0.4: the only combination whose test score did not fall, and still within noise.

Why the single changes did not transfer. Dev is YouTube comments, test is Reddit, StackExchange and Twitter 2017, and with Twitter-only training there is no test-like data to tune on. Dev gains of one or two points on 836 entities are of the same size as the seed spread we measured earlier (about 2 points), so choosing among the variants by dev is not reliable. The variant chosen by dev (V3) is significantly worse than the baseline in a paired bootstrap (-2.3, interval -4.3 to -0.2).

Why the decoding shift did not transfer. On dev the baseline has precision 66.1 and recall 42.2, so subtracting a constant from the O score raised dev F1 (delta 1.0: 54.1 against 51.5). On test the same shift moves precision from 57.2 to 46.9 but recall only from 30.9 to 33.6, so test F1 falls (39.2 against 40.1). Test entities are harder (more unseen names), so a shift tuned on the easier dev set over-predicts entities; the same happens for V3 and for the ensemble (Figure 6 in Appendix F).

What did work: the ensemble. Averaging the six models raised dev F1 from 51.5 to 52.7 and test F1 from 40.1 to 41.5, with precision 61.3 against 57.2 and recall 31.3 against 30.9. Averaging reduces the variance of each member, and the members differ in size and inputs. Two cautions. First, under the strict rule of choosing the single model with the best dev F1 the winner is V3 (53.4 against 52.7), so the ensemble was not selected by dev; we report it as a separate method fixed before its scores were seen. Second, its test gain over the baseline is +1.3 with a paired bootstrap interval of -0.2 to 2.9, which includes zero: the gain is suggestive, not proven. With the shift chosen on dev (delta 1.5) the ensemble scores 40.4, still +1.2 above the baseline with its shift.

4.4 How reliable are the differences?

With one seed, the bootstrap over test sentences gives 95% intervals of about 3 points (CRF 14.1 to 19.6, SVM 27.0 to 32.6, baseline 36.8 to 43.4, ensemble 38.0 to 44.7). The paired difference ensemble minus baseline is +1.3 (interval -0.2 to 2.9) and V3 with shift minus baseline is -2.3 (interval -4.3 to -0.2). The bootstrap does not cover training randomness: in an earlier three-seed run of the baseline architecture (train+dev protocol), seeds 42, 43 and 44 gave 42.5, 42.1, 40.2, a spread of 2.3 points, so a gain of about 2 points or less should be read as suggestive until several seeds are run. The baseline Bi-LSTM CRF leads the SVM by +10.4 (interval 7.5 to 13.1) and the SVM leads the CRF by +13.0 (interval 9.9 to 16.1). Five-fold cross-validation on train (document-grouped, before the preprocessing fix) gives 34.4 +- 4.2 for the CRF and 43.1 +- 3.3 for the SVM; these come from one domain and only show that the settings are stable.

4.5 Results by entity type

Table 5. F1 by entity type on the test set.

Type	Support	CRF	SVM	Bi-LSTM baseline	Bi-LSTM ensemble
person	429	25.9	49.7	55.0	55.9
location	150	23.7	33.8	52.0	52.9
corporation	66	0.0	4.2	17.0	14.1
product	127	2.8	8.1	13.6	15.9
creative-work	142	6.0	8.4	23.5	25.6
group	165	7.1	18.5	22.8	26.1

Person and location are easiest. Corporation, product, creative-work and group reach at most 26.1 F1 for every model: they have few training examples and overlapping definitions, and creative-work titles are long and free-form.

4.6 Ablations of the CRF and SVM

CRF features (cumulative, dev, Figure 3 left; run before the preprocessing fix): word identity alone gives 5.6 F1; shape, affixes and social-media flags add 7.9; context words and bigrams add 8.0; POS tags add 3.2; cluster IDs add 0.6 in the final setting but -1.1 at c1 = 0.3, c2 = 0.03, so their effect is small and unstable. **SVM** (dev, Figure 3 right): removing class weights raises precision to 72.2 but recall falls to 18.3 and F1 from 41.4 to 29.2; removing the GloVe vector drops F1 to 27.5. **BIO repair** (planned in Stage 1) lowered the SVM F1 on dev and on test (25.7 with repair against 29.8 without, on test), because 485 orphan I- tags in the raw output are ignored by strict scoring while repair turns them into mostly wrong entities. Repair was dropped, a decision taken on dev.

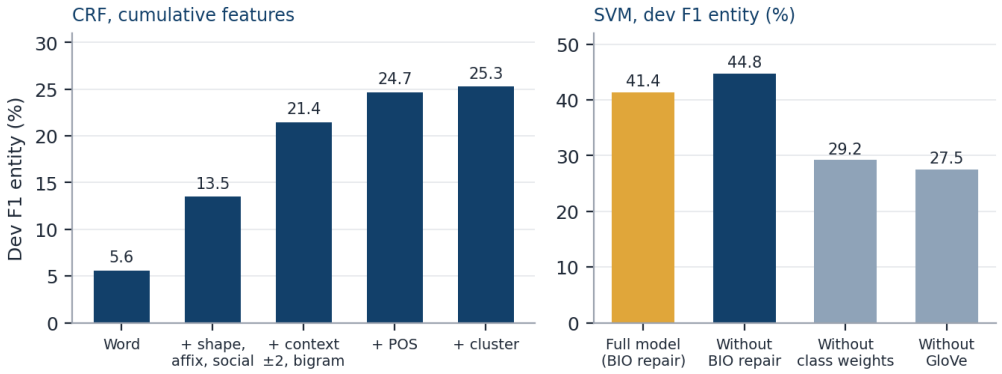


Figure 3. *Left: cumulative CRF feature ablation on dev. Right: SVM ablations on dev (amber: setting used to select C).*

4.7 Error analysis and confusion matrices

Each gold test entity falls in one category: found exactly, found with a wrong boundary or type, or missed. Predictions overlapping no gold entity are spurious.

Table 6. Outcomes for the 1,079 gold test entities.

Category	CRF	SVM	Bi-LSTM CRF (baseline)	Bi-LSTM CRF (ensemble)
Correct (exact span and type)	123	325	333	338
Wrong boundary or type	168	415	179	158
Missed entirely	788	339	567	583
Spurious predictions	106	548	68	55

The CRF misses 788 entities (73%). The SVM gets 325 right, finds 415 more with the wrong boundary or type and adds 548 spurious predictions. The baseline Bi-LSTM CRF gets 333 right and the ensemble 338, with 68 and 55 spurious predictions and 567 and 583 misses: the ensemble mainly removes spurious predictions. The confusion matrices (Figure 4) show that misses and boundary errors dominate, while type-to-type confusion is small.

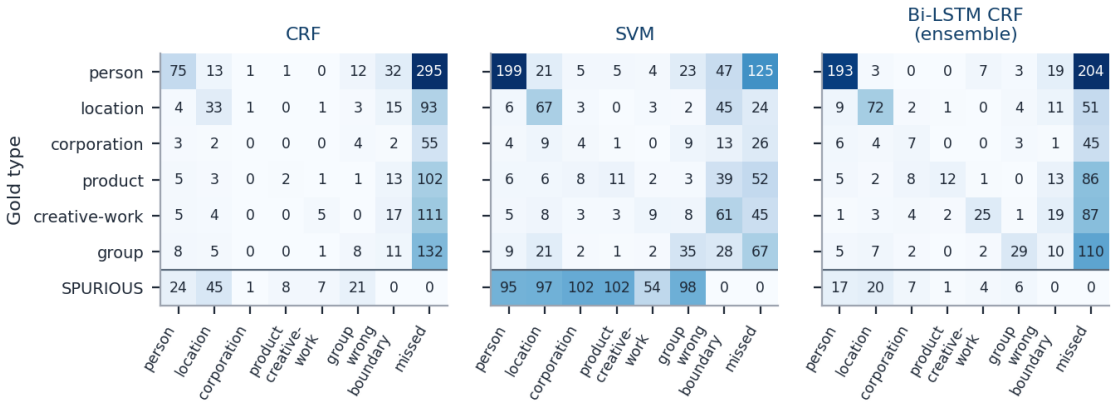


Figure 4. Entity-level confusion matrices. Rows: gold type, and SPURIOUS (predicted spans overlapping no gold entity). Columns: predicted type (exact span), wrong boundary, missed.

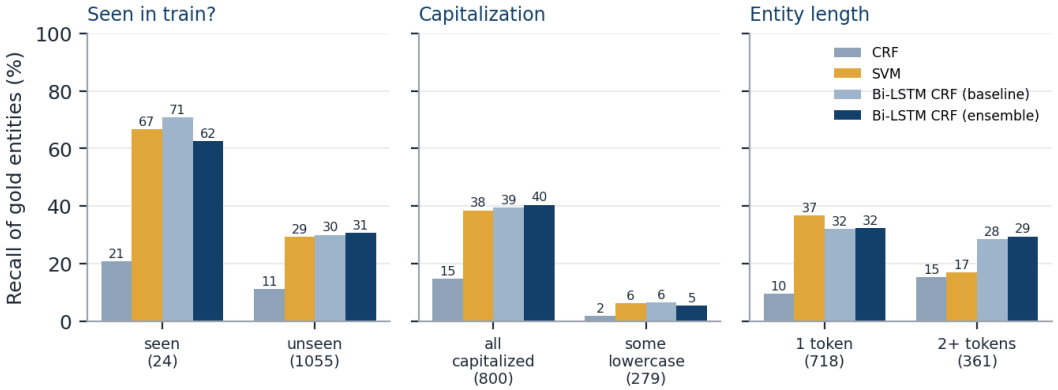


Figure 5. Recall of gold test entities by novelty (against the train file), capitalization and length (group sizes in brackets).

Only 24 of 1,079 test entities occur in the train file. On the 1055 unseen ones, recall is 11.2 (CRF), 29.3 (SVM), 30.0 (baseline) and 30.6 (ensemble). Entities with a lowercase token reach only 1.8 to 6.5%. On entities of two or more tokens the ensemble Bi-LSTM CRF recovers 29.4% against 15.2% (CRF) and 16.9% (SVM).

4.8 Examples of correct and incorrect predictions

Table 7. One correct and one incorrect test sentence per model (more in Appendix E; Bi-LSTM CRF is the improved model).

Model	Result	Sentence	Gold	Prediction
CRF	correct	* * submission title * * : Diego Rolan goal vs PSG (1 - 1)	Diego Rolan = person; PSG = group	Diego Rolan = person; PSG = group
CRF	incorrect	[From Barcelona , the stand-out youngsters are Deulofeu , Dongou and Grimaldo .	Barcelona = group; Deulofeu = person; Dongou = person; Grimaldo = person	Barcelona = location; Dongou = person; Grimaldo = person
SVM	correct	Looks like Grimaldo is linked with MU too	Grimaldo = person; MU = group	Grimaldo = person; MU = group
SVM	incorrect	[From Barcelona , the stand-out youngsters are Deulofeu , Dongou and Grimaldo .	Barcelona = group; Deulofeu = person; Dongou = person; Grimaldo = person	Barcelona = location; Deulofeu = person; Dongou = person; Grimaldo = person
Bi-LSTM CRF	correct	ITT : mutual respect between Tottenham and Chelsea fans and everyone else shitting on Giroud .	Tottenham = group; Chelsea = group; Giroud = person	Tottenham = group; Chelsea = group; Giroud = person
Bi-LSTM CRF	incorrect	[From Barcelona , the stand-out youngsters are Deulofeu , Dongou and Grimaldo .	Barcelona = group; Deulofeu = person; Dongou = person; Grimaldo = person	Barcelona = location; Deulofeu = person; Dongou = person; Grimaldo = person

5. Discussion

What worked and what did not. Each of the four single changes looked better on dev (+0.4 to +1.9) and none did better on test (-2.2 to -0.3); the decoding shift tuned on dev lowered test F1 for the baseline, V3 and the ensemble. Only the ensemble rose on both dev and test. The lesson is about tuning: when dev (YouTube) and test (Reddit, StackExchange, Twitter 2017) differ, and training uses Twitter only, small dev gains are not evidence of test gains. This is a cost of not mixing sources in training and having no test-like data for tuning.

Why the ranking of models. The SVM and Bi-LSTM CRF see dense GloVe vectors; the CRF sees GloVe only through cluster IDs and has clearly lower recall on unseen entities (11.2% against 29.3% for the SVM). In the SVM the dense vector is worth 13.9 dev F1 points. We have not tested this directly for the CRF or the Bi-LSTM CRF, so it remains a hypothesis.

Comparison with the shared task. The ensemble (41.5) and the baseline (40.1) are close to the best participant (41.86); the SVM and CRF sit lower in the reported range (26.30 to 41.86). The comparison is approximate: the released test set holds 1,079 entities after annotation fixes (the paper counts 1,040) and our surface F1 lowercases strings. With one seed and a bootstrap interval of about 3 points we do not claim to beat the shared-task system, and we do not claim that the ensemble beats the baseline beyond noise.

Limitations. One seed for every model (the ensemble members share seed 42 and differ in configuration). CRF and SVM searches, ablations and cross-validation were run before the preprocessing fix; the Bi-LSTM CRF runs are after it. Improvement choices were made on a single dev set from a different source than test. There is no run without GloVe. The neural model is written in JAX. The GloVe vector set includes words from all splits (frozen or slowly fine-tuned vectors, no labels, no training on test text).

Not done yet, on purpose. Training data are not mixed across sources. An earlier CRF-only analysis (not part of this protocol) suggested that adding YouTube data helps (test F1 16.8 with train only, 27.7 with train plus dev), and the earlier Bi-LSTM CRF run with train plus dev reached 41.6 (mean of three seeds); mixing sources is the natural next step.

6. Conclusion and Future Work

Trained on Twitter only with seed 42, the Bi-LSTM CRF baseline scores 40.1 entity F1 on the WNUT17 test set. Four changes and a decoding shift that improved dev did not improve test, and the single model chosen by dev scored 37.9. An ensemble of the six seed-42 models, defined in advance, raised test F1 to 41.5 (+1.4), close to the 41.6 obtained earlier when YouTube data were added to training, but this gain is within the noise of a single seed. The SVM (29.8) and CRF (16.8) stay lower. Unseen entities, entities with lowercase tokens and the types corporation, product, creative-work and group remain the hard cases. Next steps: run several seeds to confirm the ensemble gain, mix in sources beyond Twitter, tune on data that resemble the test domain, test the role of GloVe with a run without it, give the CRF dense embedding features, and try contextual encoders such as BERTweet.

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Appendix A. Preprocessing in Detail

- **Reading:** UTF-8; a blank line ends a document; a tab-only line ends a sentence (train: 2,394 such lines); the loader recomputes every count in Table 1.
- **Normalization** (function *normalize*): 1 HTML unescape; 2 NFKC; 3 tokens starting with http://, https://, www. or ending in .com, .net, .org, .io, .co, .ly become <URL>; 4 tokens starting with @ become <USER>; 5 a leading # is dropped; 6 every digit becomes 0; 7 a non-digit character repeated four or more times is cut to three; 8 lowercase.
- **GloVe lookup keys** (function *glove_keys*): lowercase original token; normalized form; <url> or <user> for those classes; <number> when the normalized form is only zeros; the hashtag body. 15,549 distinct keys are needed and 11,577 exist in GloVe Twitter 200d. Token coverage: train 93.4%, dev 97.7%, test 96.7%; entity tokens 94.2%, 91.0%, 84.9%.
- **POS tags:** NLTK averaged perceptron (Penn Treebank). **Clusters:** MiniBatchKMeans (k = 500, seed 42) on the 100,000 most frequent GloVe Twitter words; tokens without a vector get -1.
- **Capitalization.** The original-case token is kept for shape, is_title and is_upper features, for the capitalization embedding and for the character CNN. Lexical features, prefixes, suffixes, context words and the GloVe lookup are lowercased, and GloVe Twitter is uncased, so *Google* and *google* differ only in shape (Xx against x) and is_title for the CRF and SVM.
- **Bi-LSTM CRF:** character vocabulary from the training data (92 characters), words without a GloVe vector share one all-zero vector (trainable in V2), length buckets 8 to 128 (longest test sequence: 105 tokens, nothing truncated).

Table 8. Features used by the CRF and SVM.

Group	Features
Lexical	lowercase token, normalized form, stop-word flag (NLTK list)
Context (window +-2)	lowercase words at -2, -1, +1, +2 (BOS/EOS markers); title-case and upper-case flags at -1 and +1; left and right bigrams
Shape	collapsed shape pattern (Xxxx, dd), is_title, is_upper, has_digit, has_hyphen
Affixes	prefixes and suffixes of 1 to 4 characters (lowercase), token length (capped at 15)
Social media	is_url, is_mention, is_hashtag, is_emoji
POS tags	tag of the token and of the tokens at -1 and +1
Cluster IDs	cluster of the GloVe vector of the token and of the tokens at -1 and +1
Dense vector (SVM only)	200-d GloVe vector of the token, scaled by 0.3

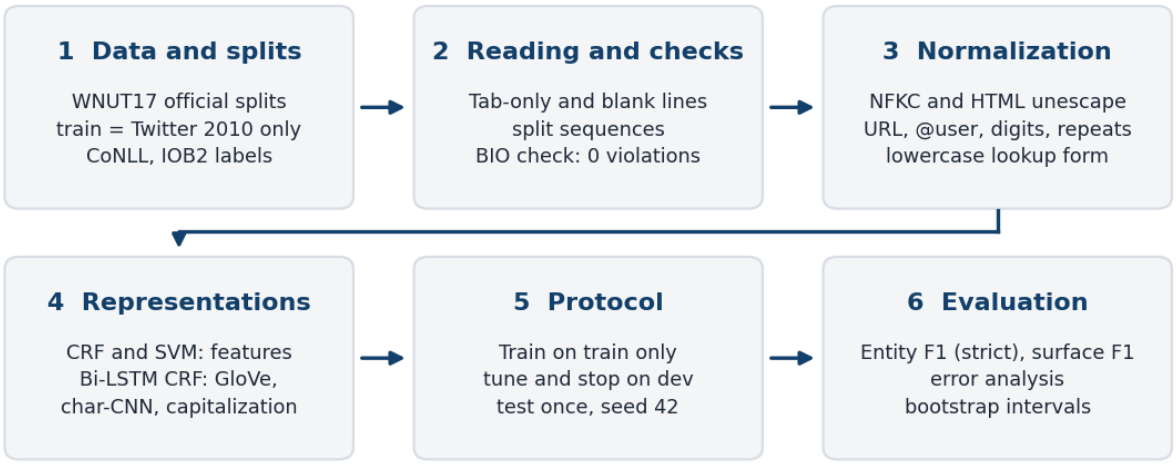


Figure 6. *Overall pipeline of the Twitter-only protocol.*

Appendix B. Deviations from the Stage 1 Plan, Protocol Changes and Bug-Fix Log

Table 9. Differences between the Stage 1 plan and this report.

Item (Stage 1 slides 21 to 25 / report)	Plan	Implemented	Status
Training data	Train on train + dev after tuning (slide 24)	Train file only (Twitter); dev only for tuning and stopping	Changed on request
Random seeds (slide 24)	Seed 42	Seed 42 for every run; no other seeds in this report	As planned
Bi-LSTM CRF improvements	Not planned	Word dropout, fine-tuned GloVe, hidden 256, POS input, ensemble, decoding shift (Section 4.3)	Added
Digit rule (slide 21)	2017 becomes 0000	Was 000 (repeat cap collapsed digit runs); now 0000	Bug fixed
URL rule (report Table 8)	http, www and *.com become <URL>	Bare domains were missing; now included	Bug fixed
Hashtag rule (report Table 8)	Drop #, process the word	# was kept in the normalized form; now dropped	Bug fixed
Fix timing	-	Bi-LSTM CRF runs use the fixed preprocessing; CRF/SVM searches, ablations and cross-validation were run before it	Disclosed
Cross-validation (slide 24)	5-fold by document on train	Train only, final settings, before the fix (CRF 34.4, SVM 43.1)	Done
SVM BIO repair (slide 22)	Apply after prediction	Built and tested, dropped: lower F1 on dev and test	Changed
SVM extras (report 3.5)	Context-window GloVe average; label history	Token GloVe vector only (x 0.3); no label history	Not done
SVM feature space	All features	Features seen at least 3 times	Changed (speed)
Context features (slide 23)	Lowercase and shape for +-2	Lowercase +-2, title/upper flags +-1, no shape for context words	Partly
Other shape flags (report 3.5)	is_lower, is_digit, has_apostrophe, is_punct, has_non_ascii, full shape	Not included	Not done
Vocabulary (slide 22)	Train vocabulary plus embedding words	All GloVe words found in any split (no labels); PAD and UNK both zero	Changed (disclosed)
Embeddings for Bi-LSTM (report 3.3)	GloVe, fine-tuned	Frozen in the baseline; fine-tuned in V2	Tested
FastText, demo tokenizer (slides 21, 22)	Alternative embedding; TweetTokenizer for new text	Not run; not built	Not done
BIOES (slide 22)	Try for Bi-LSTM CRF	One train+dev-era run, no reliable gain; not repeated here	Inconclusive
Framework	Not specified	JAX and optax (PyTorch could not be installed here)	Note

Appendix C. Chronological Log

Table 10. Work log.

#	Step	Outcome
1	Environment: Python 3.12, one CPU core (about 4 GB RAM), scikit-learn, sklearn-crfsuite, seqeval, NLTK, JAX, optax. PyTorch did not fit the free disk space.	Toolchain fixed.
2	Cloned WNUT17, recomputed all Stage 1 statistics, downloaded GloVe Twitter 200d (795 MB) and kept the needed vectors.	All counts reproduced.
3	CRF search (extended grid, 18 runs) and SVM search (C, class weights, GloVe, BIO repair) on dev.	CRF dev 25.3 (train only); SVM C = 0.03.
4	Compared the implementation with Stage 1 slides 20 to 25; found three normalization bugs and fixed them.	Bugs in Appendix B.
5	Retrained the CRF and SVM on train only; trained the Bi-LSTM CRF baseline on train only with seed 42.	Test F1 16.8, 29.8, 40.1.
6	Trained four single changes and their combination, ran a decoding-shift sweep, and averaged the six models into an ensemble; chose by dev only.	Single model chosen by dev: V3 (test 37.9); ensemble test 41.5.
7	Bootstrap, error analysis, confusion matrices, examples.	Section 4.

Appendix D. Full Tuning Grids (Dev Set, Before the Preprocessing Fix)

Table 11. CRF search (train only, 300 iterations). The selected row is shaded.

Features	c1	c2	Precision	Recall	F1 entity	F1 surface
full	0.1	0.003	53.8	15.2	23.7	23.8
full	0.1	0.01	56.1	15.3	24.1	24.2
full	0.1	0.03	58.7	15.3	24.3	24.5
full	**0.3**	**0.003**	**53.9**	**16.5**	**25.3**	**25.4**
full	0.3	0.01	55.2	15.9	24.7	24.9
full	0.3	0.03	55.6	15.3	24.0	24.4
full	1.0	0.003	54.1	14.9	23.4	23.3
full	1.0	0.01	54.7	15.3	23.9	23.9
full	1.0	0.03	56.5	15.6	24.4	24.4
no cluster	0.1	0.003	52.5	16.0	24.6	24.6
no cluster	0.1	0.01	54.6	15.6	24.2	24.2
no cluster	0.1	0.03	58.7	15.8	24.9	25.4
no cluster	0.3	0.003	52.1	16.1	24.7	24.6
no cluster	0.3	0.01	52.9	16.3	24.9	25.0
no cluster	0.3	0.03	55.5	16.3	25.2	25.3
no cluster	1.0	0.003	54.5	15.1	23.6	23.3
no cluster	1.0	0.01	54.8	15.1	23.6	23.3
no cluster	1.0	0.03	56.0	15.2	23.9	23.6

Table 12. SVM search (train only). C = 0.03 was selected with BIO repair on.

C	Dev F1 with BIO repair	Dev F1 without BIO repair
0.01	41.2	44.4
0.03	**41.4**	**44.8**
0.1	38.4	not run
0.3	37.3	not run

Appendix E. More Examples of Correct and Incorrect Predictions

Table 13. Two correct and two incorrect test sentences per model (most gold entities first, then shortest).

Model	Result	Sentence	Gold	Prediction
CRF	correct	* * submission title * * : Diego Rolan goal vs PSG (1 - 1)	Diego Rolan = person; PSG = group	Diego Rolan = person; PSG = group
CRF	correct	@ Maronti	Maronti = person	Maronti = person
CRF	incorrect	[From Barcelona , the stand-out youngsters are Deulofeu , Dongou and Grimaldo .	Barcelona = group; Deulofeu = person; Dongou = person; Grimaldo = person	Barcelona = location; Dongou = person; Grimaldo = person
CRF	incorrect	Living Computer Museum + Labs , Skyview Observatory , Smith Tower and Observation Deck ,	Living Computer Museum + Labs = location; Skyview Observatory = location; Smith Tower = location; Observation Deck = location	Skyview Observatory = corporation
SVM	correct	Looks like Grimaldo is linked with MU too	Grimaldo = person; MU = group	Grimaldo = person; MU = group
SVM	correct	81 % done with Reborn , by Jane Ederlyn https://t.co/cyEzucZWH5	Reborn = creative-work; Jane Ederlyn = person	Reborn = creative-work; Jane Ederlyn = person
SVM	incorrect	[From Barcelona , the stand-out youngsters are Deulofeu , Dongou and Grimaldo .	Barcelona = group; Deulofeu = person; Dongou = person; Grimaldo = person	Barcelona = location; Deulofeu = person; Dongou = person; Grimaldo = person
SVM	incorrect	Living Computer Museum + Labs , Skyview Observatory , Smith Tower and Observation Deck ,	Living Computer Museum + Labs = location; Skyview Observatory = location; Smith Tower = location; Observation Deck = location	Computer = product; Museum = location; + Labs = product; Skyview = corporation; Observatory = group; Smith Tower = person; Deck = person
Bi-LSTM CRF	correct	ITT : mutual respect between Tottenham and Chelsea fans and everyone else shitting on Giroud .	Tottenham = group; Chelsea = group; Giroud = person	Tottenham = group; Chelsea = group; Giroud = person
Bi-LSTM CRF	correct	Did George Smiley ' s wife Ann already know that Bill Haydon was a mole ?	George Smiley = person; Ann = person; Bill Haydon = person	George Smiley = person; Ann = person; Bill Haydon = person
Bi-LSTM CRF	incorrect	[From Barcelona , the stand-out youngsters are Deulofeu , Dongou and Grimaldo .	Barcelona = group; Deulofeu = person; Dongou = person; Grimaldo = person	Barcelona = location; Deulofeu = person; Dongou = person; Grimaldo = person
Bi-LSTM CRF	incorrect	Living Computer Museum + Labs , Skyview Observatory , Smith Tower and Observation Deck ,	Living Computer Museum + Labs = location; Skyview Observatory = location; Smith Tower = location; Observation Deck = location	Living Computer Museum = location; Skyview Observatory = location; Smith Tower = person; Observation Deck = product

Appendix F. Decoding Shift (O-bias) Sweep and Learning Curves

Table 14. O-bias sweep for V3, the single model chosen by dev. Delta is chosen on dev; the test columns are shown for transparency.

Delta	Dev P	Dev R	Dev F1	Test P	Test R	Test F1
0.0	66.8	44.5	53.4	53.2	29.4	37.9
0.5	63.7	47.0	54.1	49.2	31.5	38.4
1.0	**60.4**	**49.4**	**54.3**	**45.8**	**32.2**	**37.9**
1.5	56.8	50.1	53.2	41.8	33.8	37.4
2.0	54.2	52.3	53.2	38.4	34.8	36.5
2.5	51.0	52.6	51.8	34.7	36.0	35.3
3.0	46.7	52.6	49.5	31.5	36.6	33.9

Table 15. O-bias sweep for the ensemble (delta chosen on dev; test columns for transparency).

Delta	Dev P	Dev R	Dev F1	Test P	Test R	Test F1
0.0	68.9	42.7	52.7	61.3	31.3	41.5
0.5	66.2	45.8	54.1	56.4	33.8	42.3
1.0	64.3	49.2	55.7	51.3	34.9	41.6
1.5	**61.2**	**51.6**	**56.0**	**45.7**	**36.2**	**40.4**
2.0	57.2	52.9	55.0	40.9	37.0	38.8
2.5	52.0	53.2	52.6	35.9	37.5	36.7
3.0	47.1	53.1	49.9	31.8	37.7	34.5

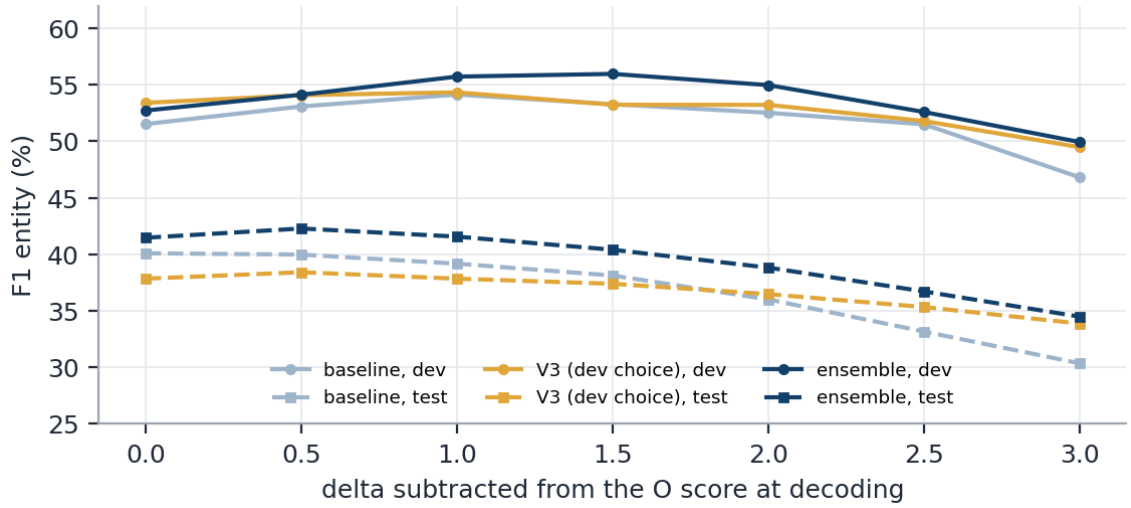


Figure 7. *Figure 6. F1 against delta on dev (solid) and test (dashed): dev prefers a shift of 1.0 to 1.5, test prefers little or none.*

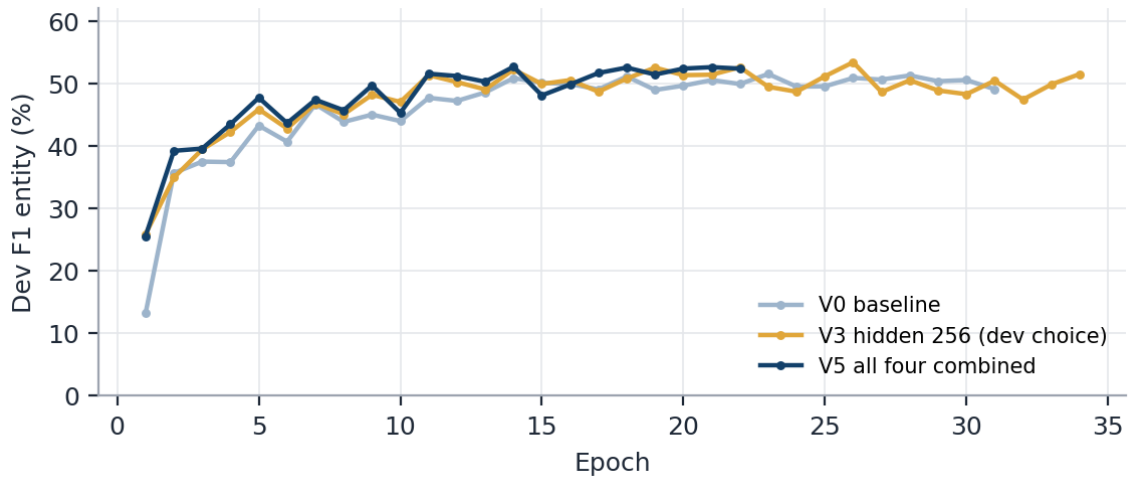


Figure 8. Dev F1 per epoch of the baseline and the final variant.

Table 16. All Bi-LSTM CRF variants (seed 42).

Step	Config	Epochs run	Best epoch	Seconds
V0 baseline (Twitter only)	{}	31	23	304
V1 word dropout 0.1	{"word_dropout": 0.1}	29	21	293
V2 fine-tuned GloVe (learning rate x0.1)	{"finetune": true, "emb_scale": 0.1}	29	21	330
V3 hidden size 256	{"hidden": 256}	34	26	469
V4 POS-tag input	{} + POS	35	31	276
V5 combination: V1 + V2 + V3 + V4	{"word_dropout": 0.1, "finetune": true, "emb_scale": 0.1, "hidden": 256} + POS	22	14	360

Appendix G. Reproducibility

Environment. Python 3.12.3; numpy 2.4.4; scipy 1.17.1; scikit-learn 1.8.0; sklearn-crfsuite 0.5.0; seqeval 1.2.2; nltk 3.10.3; jax and jaxlib 0.11.2; optax 0.2.8; matplotlib 3.10.8. One CPU core, about 4 GB RAM. A Bi-LSTM CRF epoch takes 10 to 25 seconds after a first epoch of about one minute (compilation). Every script uses seed 42.

Code (archive). The notebook reloads saved predictions and recomputes all tables and figures (set RUN_TRAINING = True to retrain). Modules: wnut_data.py, prep_glove.py, features.py, evalx.py, bilstm_crf.py. Twitter-only runs: trainonly_models.py (CRF, SVM), improve_bilstm.py (Bi-LSTM CRF variants, decoding shift) and ensemble_bilstm.py; then analysis_to.py, make_figures_to.py, build_report3.py. Search, ablation and cross-validation scripts: run_crf2.py, run_svm.py, run_svm2.py, run_cv_final.py.

Data. WNUT17 from github.com/leondz/emerging_entities_17 (CC BY 4.0); GloVe Twitter 200d from the gensim-data release glove-twitter-200. Predictions and metrics of every run are stored as JSON in the results folder.